



## Article

# Spatial Mapping of Avocado Anthracnose Severity Using UAV-Derived Vegetation Indices in Amazonas, Peru

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## Abstract

Unmanned aerial vehicle (UAV)-based multispectral remote sensing has emerged as a promising tool for monitoring crop physiological status and supporting precision disease management. However, the capacity of multispectral vegetation indices to detect foliar diseases under commercial field conditions remains insufficiently understood, particularly in perennial crops grown in humid tropical environments. This study evaluated the potential of UAV-derived multispectral vegetation indices to assess the physiological response of avocado (*Persea americana* Mill.) canopies affected by anthracnose caused by *Colletotrichum fructicola* in Amazonas, Peru. Disease incidence and severity were assessed through field evaluations, while multispectral imagery was acquired using a UAV equipped with a MicaSense RedEdge-MX Dual sensor. Vegetation indices including the Normalized Difference Vegetation Index (NDVI), Green Normalized Difference Vegetation Index (GNDVI), Soil-Adjusted Vegetation Index (SAVI), Normalized Difference Red Edge Index (NDRE), Red Edge Chlorophyll Index (CIred-edge), Plant Senescence Reflectance Index (PSRI), and Visible Atmospherically Resistant Index (VARI) were calculated, and their relationships with anthracnose incidence were analyzed using Spearman's rank correlation. Field observations confirmed a high incidence of foliar anthracnose with a heterogeneous spatial distribution across the orchard. Multispectral imagery successfully characterized spatial variability in the canopy physiological condition, revealing differences in vegetation vigor, chlorophyll-related reflectance, and senescence among experimental blocks. Nevertheless, none of the evaluated vegetation indices showed statistically significant correlations with anthracnose incidence ( $p > 0.05$ ), indicating that spectral variability primarily reflected the general canopy physiological status rather than a disease-specific spectral response. These findings demonstrate that UAV-derived multispectral vegetation indices are valuable for monitoring spatial variability in the avocado canopy condition but have limited capability for independently detecting anthracnose under humid tropical field conditions. Future studies integrating multi-temporal UAV acquisitions, hyperspectral and



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thermal imagery, LiDAR, environmental variables, and machine-learning approaches are expected to improve the early detection and spatial prediction of anthracnose in avocado production systems.

**Keywords:** *Colletotrichum fruticola*; multicriteria evaluation; UAV; severity

## 1. Introduction

Avocado (*Persea americana*) is one of the most economically and nutritionally important fruit crops worldwide, owing to its high content of monounsaturated fatty acids, dietary fiber, and bioactive compounds [1]. Over the past two decades, global avocado consumption has tripled, driven by sustained demand in international markets and its increasing recognition as a functional food [2]. This growth has created significant opportunities for producing countries, particularly in developing economies, by contributing to job creation, strengthening export capacity, and stimulating rural economies [3]. In this context, Peru has capitalized on its counter-seasonal advantage to position itself among the world's leading avocado exporters, particularly of the Hass variety, playing a strategic role in the development of the national agricultural sector [4,5]. However, the rapid expansion of avocado cultivation has also intensified pressure on natural resources, particularly water resources, posing socio-environmental challenges to the sustainability of the production system [6]. Despite these challenges, several studies have demonstrated a positive relationship between avocado exports and national economic growth, underscoring their strategic importance for national development [7].

Among the main limiting factors of avocado production, anthracnose is recognized as one of the most destructive diseases affecting both field and postharvest stages. This disease is predominantly associated with species belonging to the *Colletotrichum gloeosporioides* species complex (CGSC), a phylogenetically diverse group comprising numerous cryptic species with similar morphological characteristics and broad host ranges [8,9]. Multilocus phylogenetic analyses have shown that the *Colletotrichum gloeosporioides* species complex comprises numerous phylogenetically distinct species, including *C. fruticola*, many of which can coexist within the same agroecosystem and exhibit different levels of pathogenicity and host specificity [10]. In avocado cultivation, species of the CGSC additionally coexist with members of the *Colletotrichum acutatum* species complex and other related lineages, increasing the epidemiological complexity of anthracnose and hindering its accurate diagnosis and effective management [8].

A key epidemiological characteristic of anthracnose associated with the CGSC is the pathogen's ability to initiate infection during the preharvest stage and remain in a latent or quiescent state within host tissues until favorable conditions arise, particularly during fruit ripening and the postharvest stage [11,12]. In leaves, the disease manifests as circular or irregular necrotic lesions, typically surrounded by chlorotic halos, which reduce the photosynthetically active area and negatively affect the physiological vigor of the tree [13]. This latent infection strategy, combined with the high taxonomic diversity of the CGSC, limits the effectiveness of traditional assessment methods based on visual inspections and destructive sampling, which are constrained by limited spatial coverage, evaluator subjectivity, and low capacity for early detection [8].

Vegetation indices such as the NDVI, GNDVI, and SAVI have been widely applied to estimate canopy vigor, photosynthetic activity, chlorophyll content, and physiological stress from multispectral reflectance, supporting crop health monitoring and precision agriculture applications [14–16]. In addition, indices based on the red-edge region, such as

the Normalized Difference Red Edge Index (NDRE) and its derivatives, have shown greater sensitivity for detecting subtle physiological changes associated with fungal infections, even before visible symptom expression [17]. Several studies have reported that fungal diseases induce physiological and structural alterations in plants that are reflected in reduced vegetation index values, supporting their use as early indicators of plant health stress [16,18,19].

Despite advances in these technologies, the application of UAV-derived multispectral imagery for mapping and spectral response mapping anthracnose in avocado remains limited, particularly in tropical and Andean–Amazonian regions characterized by high relative humidity and frequent rainfall. These environmental conditions favor the emergence and spread of pathogens of the genus *Colletotrichum*, increasing the phytosanitary risk and the need for high-resolution spatial monitoring tools. Therefore, the main objective of this study was to map disease severity and the spectral response mapping of anthracnose caused by *Colletotrichum fructicola* in avocado (*Persea americana*), using vegetation indices derived from UAV-based multispectral imagery. We hypothesize that disease incidence and severity are significantly associated with variations in vegetation indices, and that red-edge-based indices exhibit greater spectral response mapping capacity than conventional indices, enabling the generation of spatial risk maps to support phytosanitary decision-making and contribute to more sustainable crop management.

## 2. Materials and Methods

### 2.1. Study Area

The study was conducted at the Magdalena Experimental Station, located in the district of Magdalena, province of Chachapoyas, Amazonas region, Peru (1939 m a.s.l.) (Figure 1). The main characteristics of the study area and the crop are presented in Table 1. In addition, preliminary assessments determined that approximately 40% of the area exhibited symptoms associated with phytosanitary problems on leaves, supporting the need to apply remote sensing tools for diagnosing and monitoring crop health status.

**Table 1.** Characteristics of the study area and the avocado crop (*Persea americana*), Hass cultivar.

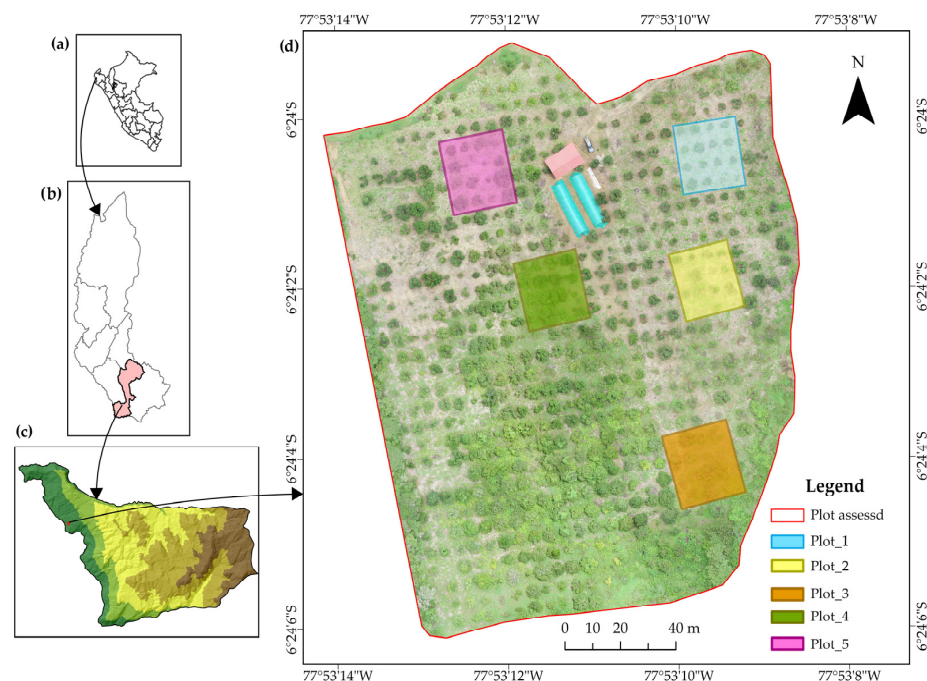
| Characteristics        | Description                                                                    |
|------------------------|--------------------------------------------------------------------------------|
| Total area of the plot | 27,549.40 m <sup>2</sup>                                                       |
| Variety                | Hass                                                                           |
| Age of the trees       | >10 years                                                                      |
| Planting system        | 5 × 5 m                                                                        |
| Density                | ~420 trees                                                                     |
| Distribution           | Five representative subplots                                                   |
| Irrigation system      | Rainfed (dependent on rainfall)                                                |
| Fertilization          | Manual, applied at critical phenological stages (growth, flowering, fruit set) |

Note. Description of the land.

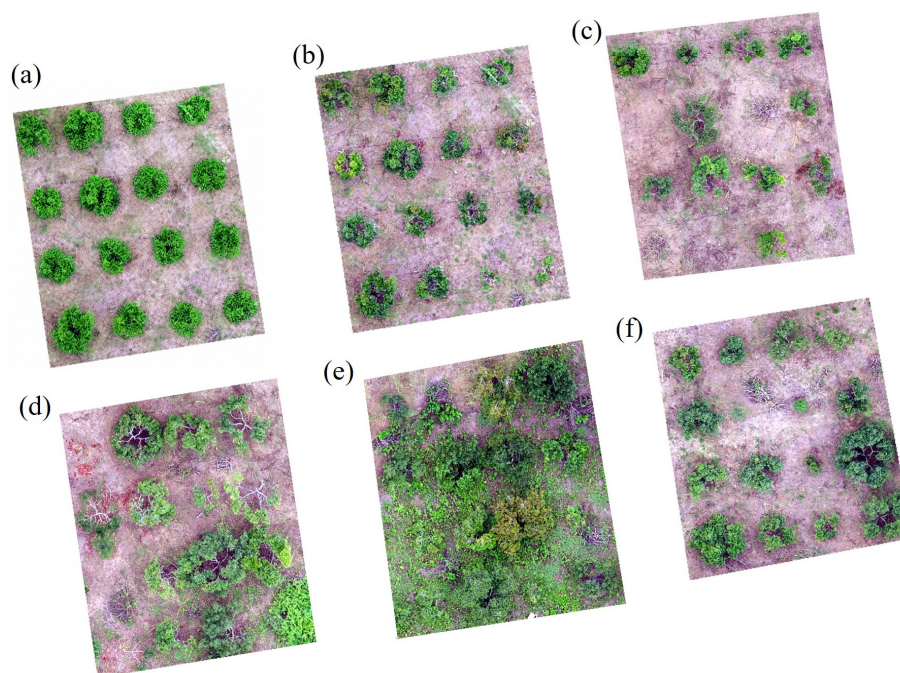
### 2.2. Sampling of Plant Material

Five plots of land with a history of anthracnose were selected [12]. In each plot, leaf samples exhibiting symptoms compatible with the disease were collected by randomly evaluating 80 plants per production unit according to the sampling design (Figure 2). Each sampling point corresponded to a plot surrounded by approximately 16 trees. A composite sample of more than 10 leaves per plot was collected, taking into account potential variability in disease distribution according to canopy position and environmental exposure. Symptom progression rates were assessed during the reproductive phase through random sampling. Leaves were cut using pruning shears disinfected with 90% alcohol and

placed in properly labeled sterile bags. Subsequently, the samples were transported to the laboratory and stored at 4 °C for analysis, where the disease incidence, severity, affected leaf area, and disease severity grade were evaluated.



**Figure 1.** Geographic location of the study area and experimental design. (a) Map showing the location of Peru in South America. (b) Location of the Amazonas Department within Peru. (c) Position of the experimental site in the Amazonas Department. (d) RGB orthomosaic of the experimental avocado orchard delineating the assessed area (red polygon) and the five experimental plots (Plots 1–5) used for data collection. Geographic coordinates (WGS 84), north orientation, and a scale bar are provided for spatial reference.



**Figure 2.** Evaluated avocado (*Persea americana*) plots affected by anthracnose caused by *Colletotrichum fructicola* under different levels of disease severity and vegetation cover. Plot (a) corresponded to the healthy control (testigo), while plots. (b–f) presented increasing incidence of anthracnose symptoms, including reduced vigor, chlorosis, defoliation, and heterogeneous canopy development.

### 2.3. Morphological Identification

#### Morphological Identification

Representative isolates (MGS03 and MGS04) associated with symptomatic avocado leaves were previously identified as *Colletotrichum fructicola* through morphological characterization, multilocus phylogenetic analyses, and pathogenicity tests conducted in the same experimental area by [20]. Briefly, isolates were cultured on PDA at  $25 \pm 2$  °C, and cultural and micromorphological characteristics were evaluated following standard taxonomic criteria for *Colletotrichum*. Molecular identification was based on concatenated analyses of four loci (ITS, CHS-1, ACT, and TUB2) using maximum likelihood inference on the CIPRES platform.

### 2.4. Assessment of Disease Incidence and Severity

The incidence of anthracnose was determined by calculating the percentage of leaves and plants that showed visible symptoms characteristic of the disease, such as circular necrotic spots, in relation to the total number of plants evaluated. This estimate was made following the methodological criteria proposed by [20].

Anthracnose severity assessments were based on the visual identification of foliar symptoms characteristic of *Colletotrichum fructicola* infection in avocado, following previously described symptomatologic criteria for avocado anthracnose [9,13] and recent observations from the same study area [21]. Evaluated symptoms included brown to dark-brown necrotic lesions, usually initiated at leaf margins or apices and frequently surrounded by chlorotic halos. As disease progressed, lesions expanded and coalesced, causing tissue dehydration, curling, and a reduction in the photosynthetically active leaf area. Under high-humidity conditions, acervuli associated with fungal sporulation were occasionally observed on symptomatic tissues. Disease severity was visually estimated according to the proportion of symptomatic tissue relative to the total leaf area.

The fungal severity per leaf was assessed using an ordinal scale ranging from 0 to 10, based on the percentage of leaf area affected by lesions ([22]). The scale used was as follows: 0 = no symptoms; 1 = 1–10%; 2 = 11–20%; 3 = 21–30%; 4 = 31–40%; 5 = 41–50%; 6 = 51–60%; 7 = 61–70%; 8 = 71–80%; 9 = 81–90%; and 10 = 91–100%. Each leaf was evaluated individually by visually distinguishing healthy (green) tissue from lesioned (brown or grayish) areas through color-based segmentation. The evaluation methodology was adapted from the criteria established by [23], and was based on the application of the following formula:

$$\text{Severity}\% = \frac{\sum(\text{area of injury})}{\text{total leaf area}} \times 100 \quad (1)$$

Finally, the Severity Index (SI) was calculated to estimate the mean disease severity, and the Area Under the Disease Progress Curve (AUDPC) was computed to quantify the temporal progression of the disease [24,25]. These metrics allowed comparisons of disease development among the evaluated treatments.

$$IS = \frac{\sum(n_i \times g_i)}{N \times g_{\max}} \times 100 \quad (2)$$

where:

$n_i$  = number of leaves in grade  $i$ ;

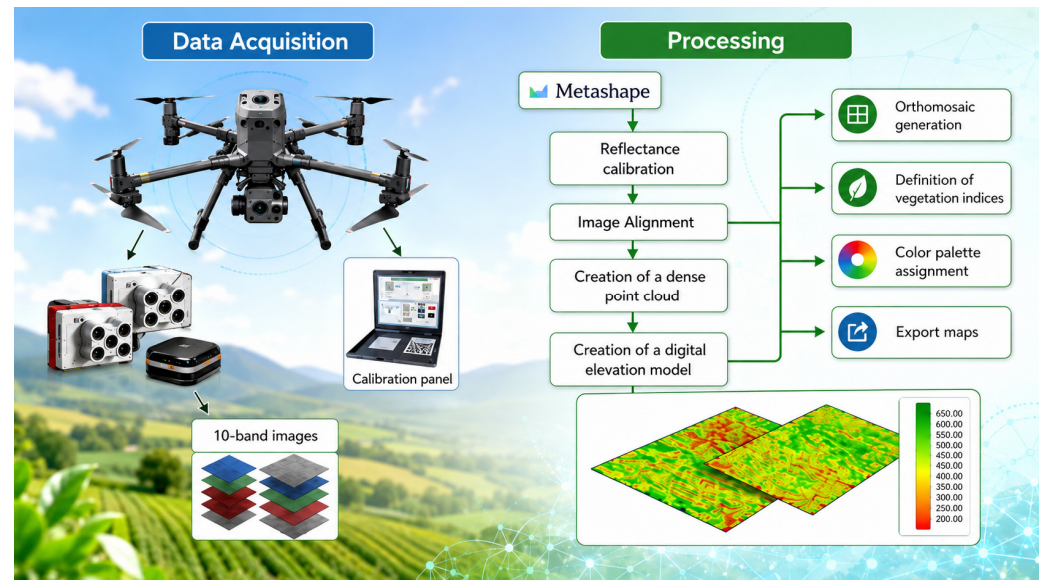
$g_i$  = value of grade  $i$  on the scale;

$N$  = total number of leaves evaluated;

$g_{\max}$  = maximum value on the scale (in this case, ten).

### 2.5. Multispectral Image Acquisition and Processing

A DJI Phantom 4 RTK UAV equipped with a high-resolution RGB camera was used [24], (Figure 3) along with the DJI Matrice 300 RTK UAV fitted with MicaSense RedEdge-MX Dual sensors (10-band multispectral). The Phantom 4 RTK incorporates a fully integrated RTK module that provides centimeter-level real-time positioning, which is essential for precision agriculture applications requiring high geospatial accuracy. Flight planning and execution were carried out using the DJI Pilot application.



**Figure 3.** The data processing workflow in Agisoft Metashape.

The orthomosaic was designed to achieve a Ground Sampling Distance (GSD) of up to 6 cm to ensure the accurate characterization of the avocado canopy. The flight parameters were set at an altitude of 75 m (RGB) and 60 m (multispectral), with a camera angle of 90°, and front and side overlaps of 80% and 80%, respectively [24,25]. Because the imagery was acquired at low flight altitude (<100 m), no atmospheric correction was applied, as atmospheric effects are considered negligible for UAV-based remote sensing under these acquisition conditions [26]. Previous studies have shown that UAV photogrammetry can achieve horizontal accuracies on the order of a few centimeters under similar flight configurations [27].

The 10-band MicaSense RedEdge-MX Dual sensor system mounted on the Matrice 300 RTK represents advanced technology for precision agriculture, capturing imagery in strategically positioned spectral bands, including Blue ( $475 \pm 32$  nm), Green ( $560 \pm 27$  nm), Red ( $668 \pm 14$  nm), Red Edge ( $717 \pm 12$  nm), NIR ( $840 \pm 40$  nm), Coastal Blue ( $444 \pm 28$  nm), Green2 ( $531 \pm 14$  nm), Red2 ( $650 \pm 16$  nm), RedEdge2 ( $705 \pm 10$  nm), and RedEdge3 ( $740 \pm 18$  nm). This configuration also covers coastal blue (444 nm), blue (475 nm), green (531 nm and 560 nm), red (650 nm and 668 nm), red edge (705 nm, 717 nm, 740 nm), and near-infrared (842 nm), enabling detailed analysis of crop phytosanitary status. This multi-spectral setup has been validated in scientific studies on crops such as maize, beans, and avocado, demonstrating its effectiveness for plant health monitoring and vigor assessment.

Orthomosaic processing was performed using the photogrammetry software Agisoft Metashape Pro (version 2.1.2), which enabled the compilation and alignment of aerial images into georeferenced orthomosaics [28], as shown in Figure 3. The processing standard was set to “medium” quality to optimize computational resources while ensuring the accurate two-dimensional reconstruction of avocado plants.

Agisoft Metashape Pro includes specific capabilities for processing multispectral images and calculating vegetation indices, and it is widely used by agricultural consultants to generate topographic maps, analyze crop status through NDVI orthomosaics, and monitor progress in precision agriculture [29].

For the extraction of multispectral data such as the NDVI, the integrated functionality of Agisoft Metashape Pro 2.1.2 was implemented, allowing the direct calculation of vegetation indices using the “Raster Calculator” tool. The process was executed through the “Set Raster Transform” option in the Tools menu (Tools > Set Raster Transform. . .), where the classical NDVI formula was specified— $(B10-B6)/(B10+B6)$ —with B10 corresponding to the near-infrared (NIR) band and B6 to the red visible band. This direct implementation in Metashape eliminates the need to export individual bands to external GIS software, enabling the automatic generation of orthomosaics with radiometrically calibrated NDVI values using reflectance panels and solar sensor information available in the image metadata. The software also allows the customization of formulas for other vegetation indices, such as the GNDVI, NDRE, or SAVI, according to the specific requirements of avocado crop analysis, taking advantage of the 10 spectral bands of the MicaSense sensor to obtain detailed information on canopy health status [30].

### 2.6. Statistical Analysis

Disease incidence among experimental blocks was analyzed using one-way analysis of variance (ANOVA), followed by Tukey’s honestly significant difference (HSD) test for multiple comparisons at a significance level of  $p < 0.05$ . Spatial distribution maps of disease incidence were generated using Kriging interpolation in QGIS.

Vegetation indices (NDVI, GNDVI, SAVI, NDRE, Cired-edge, PSRI, and VARI) were calculated from the multispectral reflectance orthomosaic. Before correlation analysis, the normality of disease incidence and each vegetation index was evaluated using the Shapiro–Wilk test. Because disease incidence and most vegetation indices did not satisfy the normality assumption ( $p < 0.05$ ), the relationship between vegetation indices and disease incidence was assessed using Spearman’s rank correlation coefficient ( $\rho$ ), a non-parametric method appropriate for non-normally distributed data and monotonic relationships. Correlation coefficients,  $p$ -values, and 95% confidence intervals were calculated for each vegetation index.

## 3. Results

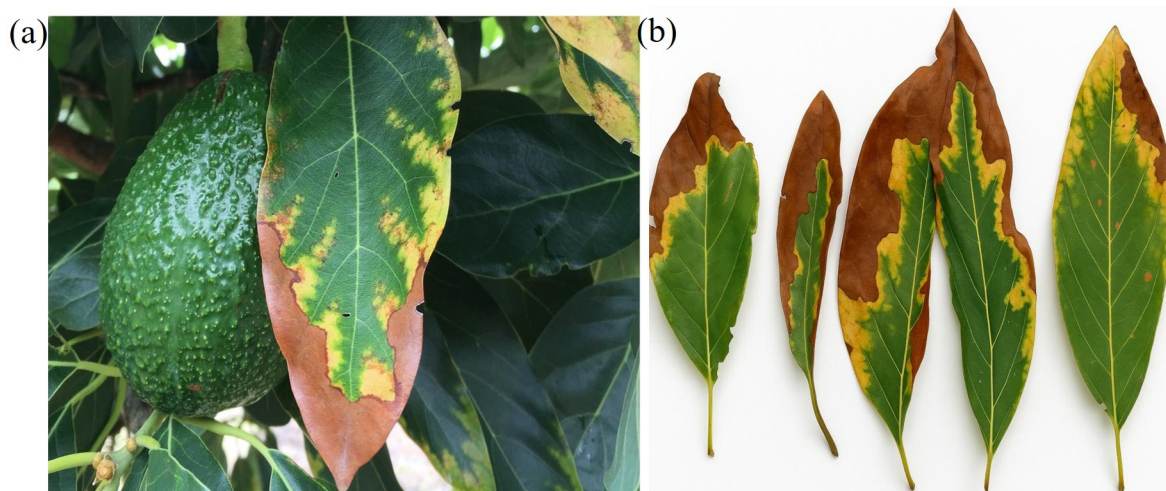
### 3.1. Field Observations and Symptoms of the Disease

Foliar infection by *Colletotrichum fructicola* in avocado begins as small necrotic spots ranging from brown to dark brown, typically appearing on leaf margins or tips (Figure 4). In the early stages, these lesions are often surrounded by chlorotic halos and expand over time. As the disease progresses, spots enlarge and merge, leading to extensive necrosis that reduces the photosynthetic area. In severe cases, leaves exhibit overall darkening, loss of turgor, curling, and tissue dehydration. Under humid conditions, fungal acervuli may be observed on the lesion surface. This symptom pattern is typical of anthracnose and can result in significant yield losses.

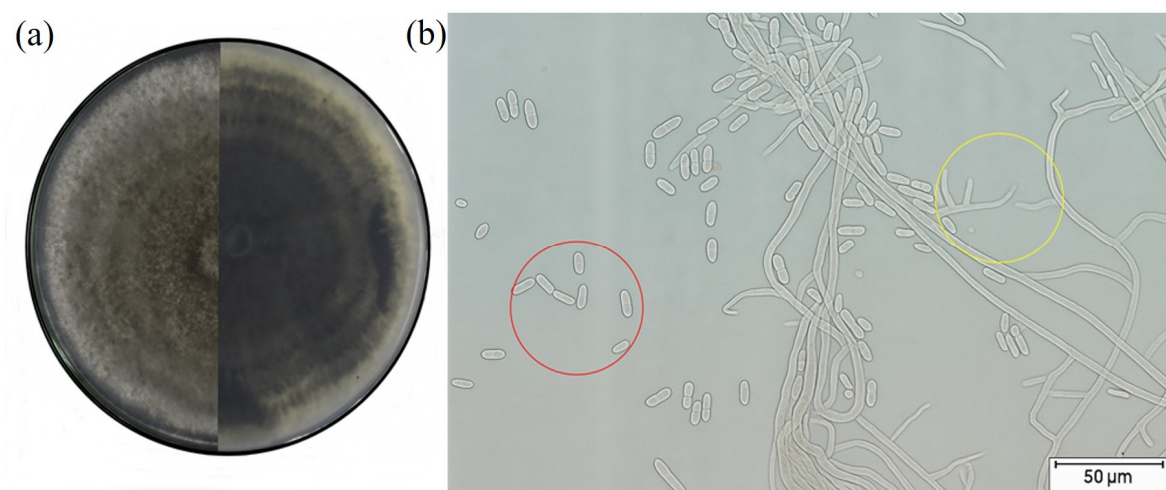
### 3.2. Morphological–Molecular Identification

The causal agent associated with symptomatic avocado leaves was identified as *Colletotrichum fructicola* through routine morphological and molecular analyses conducted at the Plant Health Laboratory of the Universidad Nacional Toribio Rodríguez de Mendoza. Isolates showed rapid growth on PDA, producing dense, velvety colonies with concentric zonation and dark pigmentation (Figure 5a). Microscopic examination revealed hyaline,

septate conidiophores and cylindrical to clavate conidia (Figure 5b). Molecular confirmation was performed using four genetic markers on representative isolates (MGS03 and MGS04), supporting their assignment to *C. fruticola*.



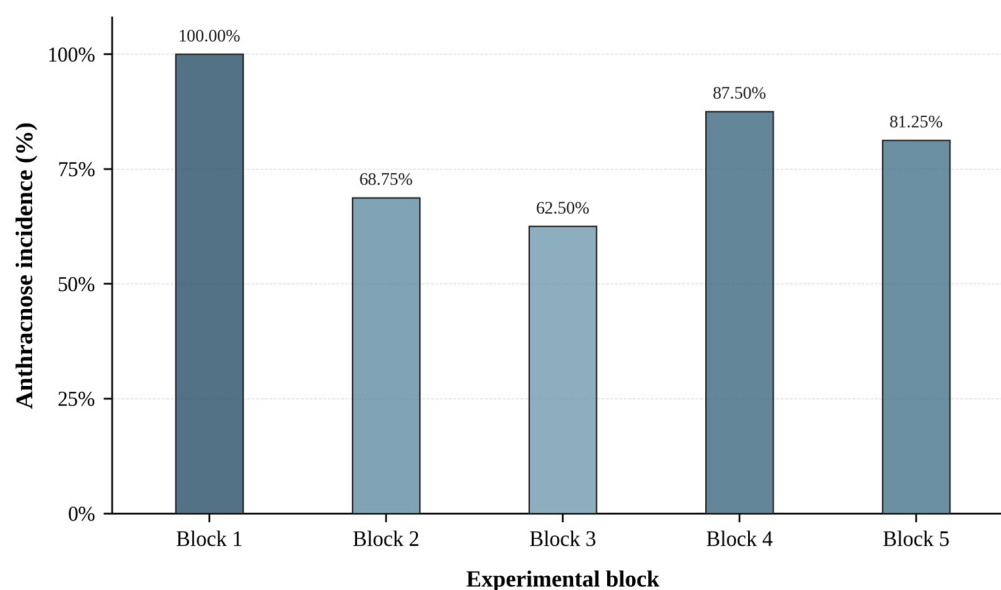
**Figure 4.** Visual symptoms and signs of anthracnose in avocado (*Persea americana*) caused by *Colletotrichum fruticola* under field and laboratory conditions. (a) Avocado leaves in the field showing characteristic symptoms of marginal necrosis, chlorosis, and lesions associated with *C. fruticola* infection. (b) Leaves collected and evaluated under laboratory conditions, showing different levels of disease severity and progression of foliar damage.



**Figure 5.** Morphological characteristics of *Colletotrichum fruticola* isolated from avocado leaves affected by anthracnose. (a) Upper and reverse colony views on PDA after seven days at 28 °C, showing dense mycelial growth and dark pigmentation. (b) Microscopic observation of hyaline conidiophores and cylindrical conidia (red circle), with branched hyphal structures indicated in the yellow circle; scale bar = 50 µm.

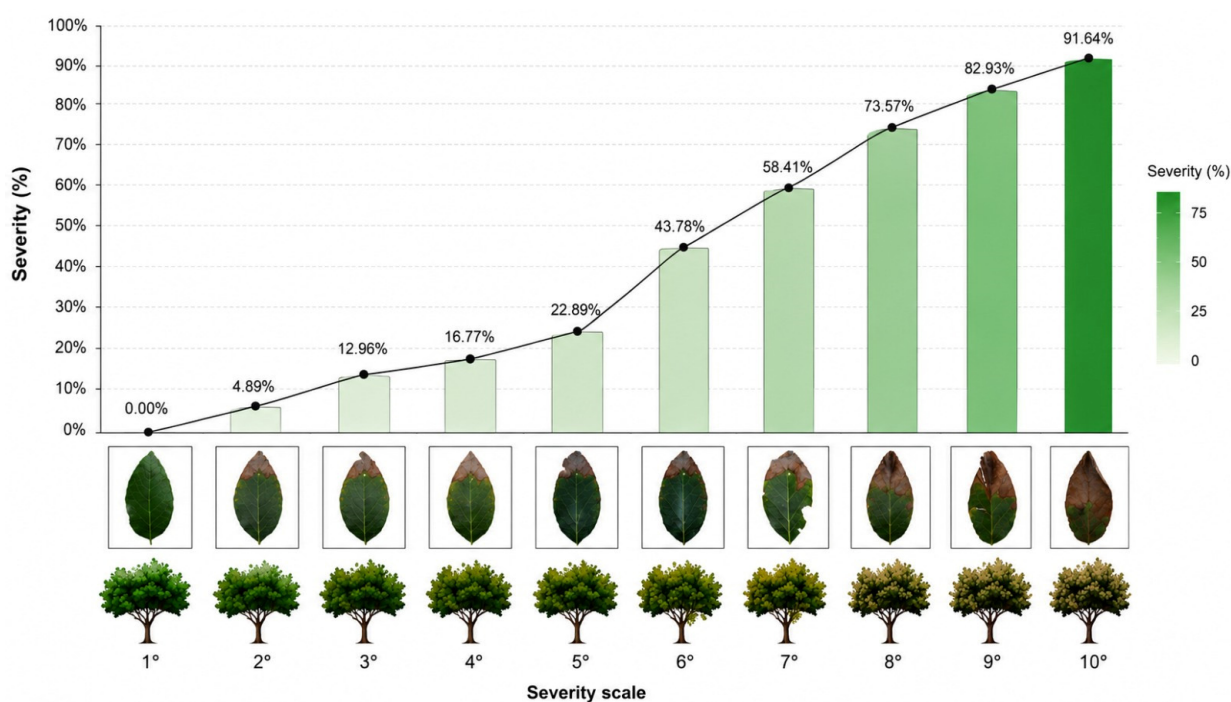
### 3.3. Assessment of Anthracnose Incidence and Severity

The incidence of *C. fruticola* differed markedly among the evaluated blocks (Figure 6). Block 1 exhibited the highest incidence, reaching 100%, indicating widespread infection in the crop. In contrast, Blocks 2 and 3 showed the lowest incidences, with 68.75% and 62.50%, respectively, suggesting relatively lower disease severity. Block 4 displayed a high incidence (87.50%), while Block 5 presented an intermediate–high value (81.25%). Overall, the trend reflects a decrease in incidence from Block 1 to Block 3, followed by an increase in Blocks 4 and 5. These results confirm a heterogeneous spatial distribution of anthracnose within the study area.



**Figure 6.** Anthracnose incidence varied among the five experimental blocks, ranging from 62.50% to 100.00%. Block 1 showed the highest incidence (100.00%), followed by Blocks 4 (87.50%) and 5 (81.25%). The lowest incidence was recorded in Block 3 (62.50%), followed by Block 2 (68.75%). Overall, the mean incidence was 80.0%, indicating a heterogeneous spatial distribution of the disease.

The graph shows a clear and gradual progression of foliar anthracnose severity, ranging from negligible or very low levels to high values along the assessment scale (Figure 7). The initial grades (1–3) exhibited severities below 15%, corresponding to early lesions with limited leaf coverage. From grade 4 onward, a steady increase in severity was observed, reaching intermediate values between 22.8% and 58.4% in grades 5–7, reflecting a significant expansion of necrotic lesions on the leaves.



**Figure 7.** Severity gradient of foliar anthracnose in avocado (*Persea americana*) leaves caused by *Colletotrichum fruticola*, ranging from healthy leaves (0.00%) to severely affected leaves (91.64%). The severity scale was established based on the proportion of diseased foliar area, showing progressive tissue necrosis and chlorosis associated with increasing disease development.

In the most advanced grades (8–10), severity exceeded 70%, reaching a maximum of 91.64%, indicating severe leaf damage and substantial loss of photosynthetic tissue. This pattern confirms the ability of the visual scale to discriminate among different levels of anthracnose impact and its correspondence with the progressive increase in damage observed at both the leaf and whole-plant levels. These results support the use of this classification for quantifying disease severity in field assessments.

Disease incidence did not show statistically significant differences among plots according to analysis of variance (ANOVA;  $F = 2.35$ ;  $p = 0.0619$ ). Consistently, Tukey's multiple comparison test ( $p < 0.05$ ) did not detect differences among the mean incidence values of the evaluated plot, which were grouped into a single statistical group. Similarly, severity analysis revealed no significant differences among plots (ANOVA;  $F = 1.63$ ;  $p = 0.1751$ ). Tukey's test confirmed this result, as all plots shared the same statistical letter, indicating comparable severity among blocks. Overall, these results suggest a relatively homogeneous distribution of the disease in terms of both incidence and severity at the plot level.

### 3.4. Vegetation Index Identification

Spatial analysis of vegetation indices derived from UAV-based multispectral imagery revealed marked spatial variability in canopy reflectance among the five experimental blocks (Figures 8 and 9). The NDVI, GNDVI, SAVI, NDRE, CIred-edge, and PSRI maps showed heterogeneous spatial patterns, indicating differences in the canopy physiological condition across the study area. Lower values of the NDVI, GNDVI, SAVI, and NDRE were observed in localized portions of several blocks, whereas higher values were distributed in other areas, reflecting differences in vegetation vigor and chlorophyll-related spectral responses.

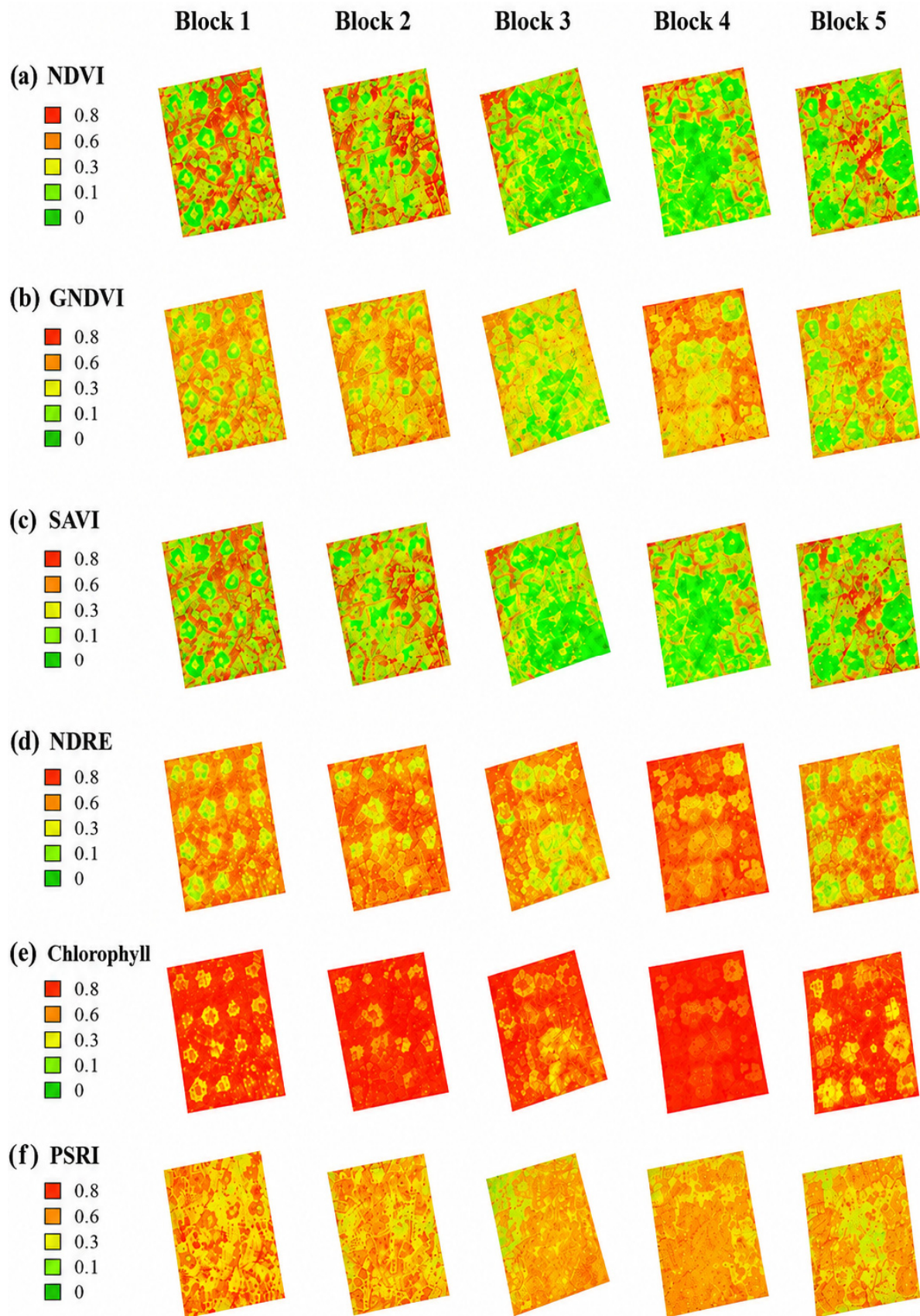
The CIred-edge and PSRI indices also exhibited spatial variability throughout the orchard. CIred-edge highlighted differences in chlorophyll-related reflectance, whereas the PSRI showed localized increases that may be associated with changes in leaf senescence or physiological stress. However, these spatial patterns should be interpreted as indicators of the general canopy condition rather than direct evidence of anthracnose infection.

Although the vegetation indices revealed spatial heterogeneity within the orchard, subsequent correlation analysis indicated that none of the evaluated indices was significantly associated with disease incidence (Spearman,  $p > 0.05$ ). These results suggest that the spatial variability observed in canopy reflectance was likely influenced by multiple biotic and abiotic factors and that a single multispectral UAV acquisition was insufficient to establish a robust relationship between vegetation indices and anthracnose incidence under the evaluated field conditions.

Analysis of vegetation indices derived from UAV-acquired multispectral imagery revealed considerable spatial variability in canopy spectral responses across the experimental avocado orchard. The NDVI, GNDVI, and SAVI maps showed predominantly medium-to-high values, indicating overall canopy vigor, with localized areas exhibiting lower values that may reflect variations in the plant physiological condition. Similarly, the NDRE and CIred-edge revealed spatial differences in chlorophyll-related reflectance, whereas the PSRI exhibited localized increases associated with canopy senescence or physiological stress.

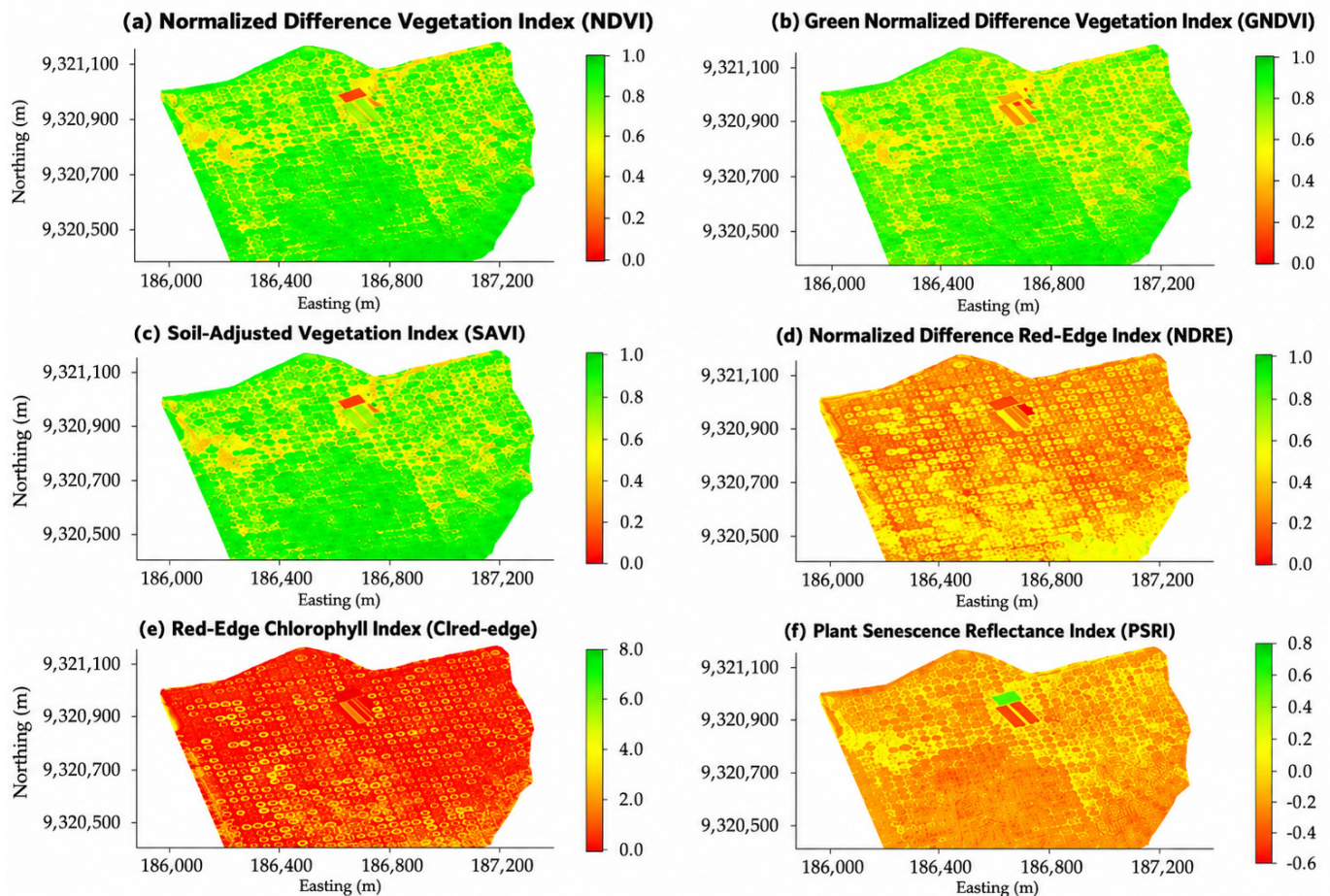
Although these indices highlighted heterogeneous spatial patterns within the orchard, the subsequent Spearman correlation analysis showed that none of the evaluated vegetation indices was significantly associated with anthracnose incidence ( $p > 0.05$ ). Therefore, the observed spatial variability should be interpreted as differences in the general canopy physiological condition rather than as a direct spectral signature of *Colletotrichum fructicola* infection. These findings suggest that, under the conditions evaluated and using a

single UAV acquisition, multispectral vegetation indices alone were insufficient to reliably discriminate anthracnose incidence.



**Figure 8.** The spatial distribution of the NDVI, GNDVI, SAVI, NDRE, Chlorophyll, and PSRI indices across five experimental blocks. Variability in plant vigor is observed among blocks, with Blocks 3

and 4 showing greater contrasts in stress levels and lower uniformity. Chlorophyll-related indices (NDRE and Chlorophyll) provide a clearer physiological differentiation of the crop. Overall, the figure reveals spatial heterogeneity in canopy health conditions. (a) NDVI, showing spatial variability in canopy vigor, with generally higher values in greener areas and lower values in yellow–red areas; (b) GNDVI, illustrating differences in green vegetation density and vigor among the blocks, with particularly pronounced spatial variation in Blocks 3 and 4; (c) SAVI, showing patterns of vegetation vigor similar to NDVI while accounting for soil-background effects; (d) NDRE, revealing spatial variation in chlorophyll-related vegetation status and highlighting areas with contrasting crop physiological conditions; (e) Chlorophyll index, showing the spatial distribution of estimated canopy chlorophyll content and providing clear differentiation of physiological variability among blocks; and (f) PSRI, indicating spatial variation associated with plant senescence and stress.



**Figure 9.** Spatial distribution of vegetation indices (NDVI, GNDVI, SAVI, NDRE, Clred-edge, and PSRI) derived from UAV-acquired multispectral imagery in the experimental avocado orchard. The maps illustrate the spatial variability of canopy spectral responses across the study area. Although localized differences in vegetation vigor and physiological condition were observed, subsequent Spearman correlation analysis revealed no significant relationship between vegetation indices and anthracnose incidence ( $p > 0.05$ ). (a) NDVI: Distribution of the Normalized Difference Vegetation Index, reflecting overall canopy vigor. (b) GNDVI: Distribution of the Green Normalized Difference Vegetation Index, associated with chlorophyll content and physiological condition. (c) SAVI: Soil-Adjusted Vegetation Index, which minimizes the influence of soil background. (d) NDRE: Normalized Difference Red Edge Index, sensitive to variations in chlorophyll content. (e) Clred-edge: Red Edge Chlorophyll Index, related to chlorophyll concentration and canopy physiological status. (f) PSRI: Plant Senescence Reflectance Index, associated with senescence processes and chlorophyll degradation.

### 3.5. Correlation Between Vegetation Indices and Anthracnose Incidence

To evaluate the relationship between vegetation indices and anthracnose incidence, Spearman's rank correlation analysis was performed after verifying that disease incidence and most vegetation indices did not meet the normality assumption according to the Shapiro–Wilk test. The correlation coefficients, corresponding  $p$ -values, and 95% confidence intervals are presented in Table 2.

**Table 2.** Spearman rank correlation coefficients ( $\rho$ ) between vegetation indices and anthracnose incidence in avocado trees.

| Vegetation Index | Spearman's $\rho$ | 95% CI          | $p$ -Value |
|------------------|-------------------|-----------------|------------|
| PSRI             | −0.102            | −0.334 to 0.138 | 0.394      |
| CIred-edge       | −0.054            | −0.269 to 0.171 | 0.653      |
| NDRE             | −0.05             | −0.274 to 0.185 | 0.675      |
| GNDVI            | −0.025            | −0.238 to 0.193 | 0.834      |
| NDVI             | 0.094             | −0.140 to 0.322 | 0.433      |
| SAVI             | 0.097             | −0.149 to 0.333 | 0.418      |
| VARI             | 0.154             | −0.093 to 0.372 | 0.196      |

Note: Spearman's rank correlation coefficient ( $\rho$ ) was used because disease incidence and most vegetation indices did not satisfy the normality assumption according to the Shapiro–Wilk test ( $p < 0.05$ ).

The results showed weak correlations between anthracnose incidence and all evaluated vegetation indices (Table 2). The correlation coefficients ( $\rho$ ) ranged from −0.102 for the PSRI to 0.154 for the VARI. The PSRI, CIred-edge, NDRE, and GNDVI exhibited weak negative correlations with disease incidence, whereas the NDVI, SAVI, and VARI showed weak positive correlations. However, none of these relationships was statistically significant ( $p > 0.05$ ), and all the 95% confidence intervals included zero, indicating the absence of a significant monotonic relationship between vegetation indices and anthracnose incidence under the evaluated conditions.

These findings indicate that, although the UAV-derived multispectral imagery revealed spatial variability in canopy spectral responses, the evaluated vegetation indices were not significantly associated with anthracnose incidence. Therefore, the observed spectral variability likely reflects general differences in the canopy physiological condition rather than a direct response to disease infection.

#### 3.5.1. Analysis of Canopy Spectral Variability at the Plant Level Using Vegetation Indices

Vegetation indices derived from UAV-acquired multispectral imagery revealed spatial variability in canopy spectral responses across the five experimental plots (Figures 8 and 9). The evaluated indices (NDVI, GNDVI, SAVI, NDRE, CIred-edge, PSRI, and VARI) exhibited differences in reflectance values among individual tree crowns, indicating heterogeneity in the canopy physiological condition throughout the orchard.

The NDVI, GNDVI, and SAVI maps showed predominantly medium-to-high values, suggesting relatively uniform vegetation vigor across most of the evaluated area, although localized sectors with lower values were observed. Likewise, the NDRE and CIred-edge revealed spatial variation in chlorophyll-related reflectance, whereas the PSRI displayed localized increases associated with differences in canopy senescence or physiological condition. Overall, the vegetation index maps highlighted heterogeneous spatial patterns in canopy reflectance among the experimental plots.

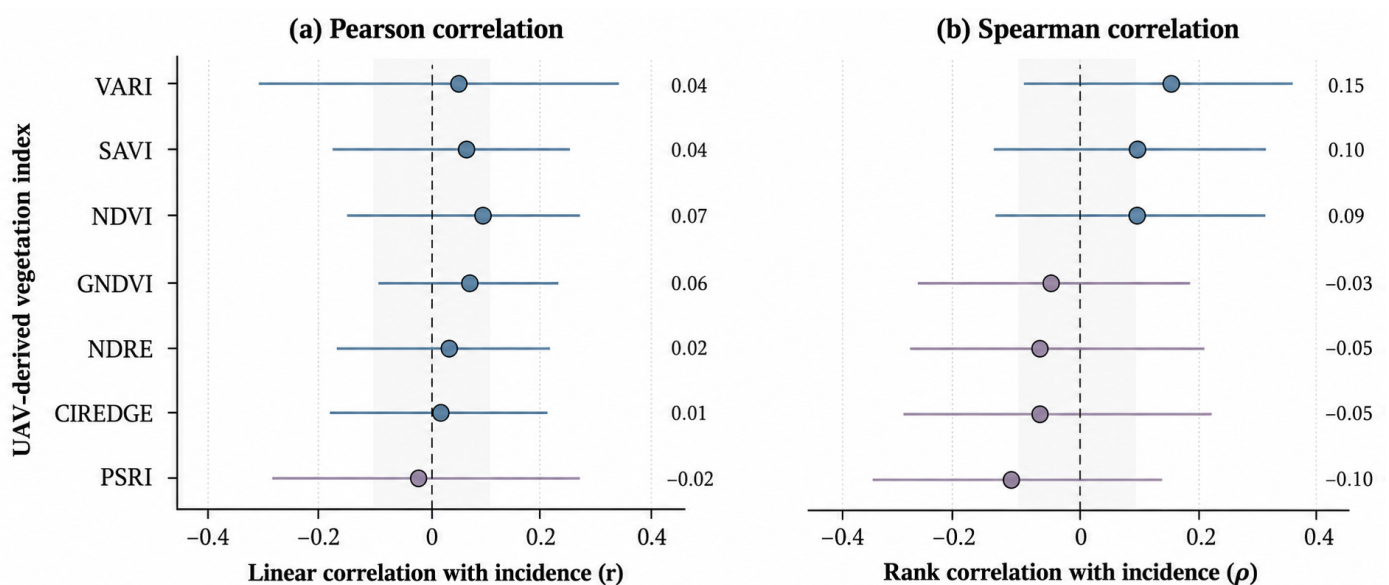
To determine whether these spectral variations were associated with anthracnose incidence, a Spearman rank correlation analysis was performed after confirming the non-normal distribution of disease incidence and most vegetation indices using the Shapiro–

Wilk test. The results showed weak correlations for all the evaluated indices, with the correlation coefficients ranging from  $-0.102$  (PSRI) to  $0.154$  (VARI) (Table 2). The PSRI, CIred-edge, NDRE, and GNDVI exhibited weak negative correlations with disease incidence, whereas the NDVI, SAVI, and VARI showed weak positive correlations. However, none of the evaluated vegetation indices showed statistically significant relationships with anthracnose incidence ( $p > 0.05$ ), and all the 95% confidence intervals included zero.

These findings indicate that, although UAV-derived multispectral imagery successfully characterized the spatial variability in the canopy physiological condition, the evaluated vegetation indices were not significantly associated with anthracnose incidence under the conditions of this study. Therefore, the observed spectral variability should be interpreted as reflecting general differences in canopy condition rather than a direct spectral response to *Colletotrichum fructicola* infection.

### 3.5.2. Relationship Between Anthracnose Incidence and Vegetation Indices

To evaluate the relationship between vegetation indices and anthracnose incidence, Spearman's rank correlation analysis was performed after verifying the non-normal distribution of disease incidence and most vegetation indices using the Shapiro–Wilk test. The correlation coefficients, corresponding  $p$ -values, and 95% confidence intervals are presented in Table 2, while Figure 10 provides a graphical comparison of the Pearson and Spearman correlation coefficients.



**Figure 10.** Pearson (a) and Spearman (b) correlation coefficients between UAV-derived vegetation indices and anthracnose incidence. Points represent correlation coefficients, and horizontal bars indicate the corresponding 95% confidence intervals. The vertical dashed line indicates zero correlation. None of the evaluated vegetation indices showed statistically significant correlations with anthracnose incidence, as all the confidence intervals overlapped zero.

The Spearman correlation coefficients ranged from  $-0.102$  (PSRI) to  $0.154$  (VARI). The PSRI, CIred-edge, NDRE, and GNDVI exhibited weak negative correlations with disease incidence, whereas the NDVI, SAVI, and VARI showed weak positive correlations. However, none of the evaluated vegetation indices showed statistically significant relationships with anthracnose incidence ( $p > 0.05$ ), and all the 95% confidence intervals included zero.

Pearson and Spearman analyses produced similar patterns, with the correlation coefficients remaining close to zero for all vegetation indices. This consistency indicates the absence of both linear and monotonic relationships between UAV-derived vegeta-

tion indices and anthracnose incidence under the evaluated conditions. Consequently, although multispectral imagery captured spatial variability in the canopy physiological condition, the evaluated vegetation indices alone were insufficient to reliably discriminate anthracnose incidence.

### 3.6. Spectral Response of Vegetation Indices to Incidence of Anthracnose (*Colletotrichum fructicola*) in Avocado (*Persea americana*)

The spatial distribution maps of the NDVI, GNDVI, SAVI, NDRE, Cired-edge, PSRI, and VARI revealed heterogeneous canopy spectral responses across the experimental avocado orchard. Localized variations in vegetation vigor, chlorophyll-related reflectance, and canopy physiological condition were observed among the evaluated plots, indicating spatial heterogeneity in the orchard.

Despite these spatial differences, the correlation analysis demonstrated that none of the evaluated vegetation indices was significantly associated with anthracnose incidence. These findings suggest that the observed spectral variability may have been influenced by multiple factors affecting canopy physiology, including natural plant variability, environmental conditions, nutritional status, or other biotic and abiotic stresses, rather than exclusively by *Colletotrichum fructicola* infection.

Consequently, under the conditions of this study, multispectral vegetation indices derived from a single UAV acquisition were useful for characterizing spatial variability in the canopy condition but were not sufficient to reliably detect anthracnose incidence. Future studies should incorporate multi-temporal UAV acquisitions throughout disease development and integrate complementary sensing technologies, such as thermal, hyperspectral, or LiDAR data, to improve disease detection and predictive modeling.

## 4. Discussion

This study evaluated the potential of UAV-derived multispectral vegetation indices to characterize canopy physiological variability and assess their relationship with the incidence of anthracnose caused by *Colletotrichum fructicola* in avocado (*Persea americana*) under humid tropical field conditions. The multispectral orthomosaics successfully revealed marked spatial heterogeneity in the canopy condition across the orchard, indicating differences in vegetation vigor, chlorophyll-related reflectance, pigment dynamics, and senescence. However, none of the evaluated vegetation indices, including the NDVI, GNDVI, SAVI, NDRE, Cired-edge, PSRI, and VARI, showed statistically significant correlations with anthracnose incidence. This result suggests that the spectral variability captured from a single UAV acquisition mainly reflected general canopy physiological status rather than a disease-specific spectral response. Similar limitations have been reported in UAV-based disease-monitoring studies, where vegetation indices have been shown to detect crop stress patterns but not always discriminate specific pathogens under complex field conditions [31,32], while more recent UAV studies have reported comparable findings under field conditions [16,33].

Several biological and methodological factors may explain the absence of significant relationships between vegetation indices and anthracnose incidence. Anthracnose initially develops as localized foliar lesions that occupy only a small fraction of the leaf surface, whereas canopy-level vegetation indices represent the average reflectance of the entire tree crown [32]. Consequently, disease-induced spectral changes may be diluted by the reflectance of healthy foliage, particularly during the early stages of infection. Furthermore, canopy reflectance is simultaneously influenced by the canopy architecture, leaf age, chlorophyll concentration, water availability, nutrient status, soil moisture, illumination conditions, and atmospheric variability, all of which may obscure pathogen-induced

spectral responses [34–37]. Recent evidence also indicates that soil moisture and environmental variability substantially influence canopy reflectance independently of disease development [38].

The spectral behavior observed in this study is consistent with the physiological characteristics of dense avocado canopies. Traditional vegetation indices such as the NDVI, GNDVI, and SAVI are known to saturate under conditions of high biomass and elevated chlorophyll content, thereby reducing their sensitivity to subtle physiological disturbances [34,37]. In contrast, indices based on the red-edge spectral region, including the NDRE and CIred-edge, are generally more responsive to variations in chlorophyll concentration and leaf internal structure because the red-edge wavelength is highly sensitive to early physiological stress [35,36]. Nevertheless, despite their recognized physiological sensitivity, neither the NDRE nor CIred-edge showed statistically significant correlations with anthracnose incidence in the present study, suggesting that environmental and physiological variability outweighed disease-related spectral responses.

Previous studies have reported stronger relationships between UAV-derived spectral information and plant diseases when disease monitoring incorporated repeated image acquisitions or complementary sensing technologies. Thermal infrared imagery allows the early detection of changes in canopy temperature associated with pathogen infection before visual symptoms become evident [31,32]. Likewise, hyperspectral sensors provide hundreds of contiguous spectral bands capable of detecting subtle biochemical alterations in pigments, water content, and secondary metabolites [39], whereas LiDAR contributes structural information describing canopy architecture and foliage distribution, improving disease discrimination when integrated with spectral information [16]. Consequently, multisensor approaches generally outperform multispectral vegetation indices alone for disease detection [33].

An important implication of the present findings is that vegetation indices should not be interpreted as pathogen-specific indicators. The NDVI, GNDVI, SAVI, NDRE, CIred-edge, PSRI, and VARI primarily respond to the chlorophyll concentration, pigment composition, vegetation vigor, canopy density, and senescence [34,37]. These physiological characteristics are influenced by multiple stress factors, including water deficit, nutritional imbalance, excessive irradiance, natural aging, and pathogen infection [32]. Consequently, multispectral vegetation indices provide reliable indicators of generalized plant stress but cannot independently distinguish anthracnose from other sources of physiological variability [16,31,33].

The present study also highlights several methodological limitations. Disease assessment was based on a single UAV multispectral acquisition, representing only one stage of anthracnose development. Because disease progression is dynamic, repeated image acquisitions throughout the growing season would likely capture temporal physiological responses more effectively [32]. Furthermore, only multispectral vegetation indices were evaluated, providing a relatively limited spectral representation compared with hyperspectral, thermal infrared, fluorescence, or LiDAR sensors [31,39]. Differences among studies may also reflect variations in spectral resolution, disease severity, acquisition timing, environmental conditions, image-processing workflows, and statistical methodologies [16].

The findings demonstrate that UAV-derived multispectral vegetation indices are valuable tools for describing the spatial variability of the avocado canopy physiological condition but, when used independently and from a single acquisition, have limited capability for reliably discriminating anthracnose incidence under humid tropical conditions. Rather than reducing the value of UAV remote sensing, these results define its current limitations and provide a foundation for improving future disease-monitoring strategies. Future research should incorporate multi-temporal UAV surveys, hyperspectral and ther-

mal imagery, terrestrial or UAV-based LiDAR, environmental variables, and advanced machine-learning algorithms such as Random Forest, Support Vector Machines, and deep learning models to improve the early detection and spatial prediction of anthracnose in avocado production systems.

## 5. Conclusions

Leaf anthracnose caused by *Colletotrichum fructicola* exhibited high incidence and a heterogeneous spatial distribution in avocado orchards in the Amazonas department of Peru. UAV-derived multispectral imagery successfully characterized the spatial variability of the canopy physiological condition through vegetation indices, revealing differences in vegetation vigor and chlorophyll-related responses across the orchard. However, none of the evaluated vegetation indices (NDVI, GNDVI, SAVI, NDRE, C<sub>I</sub>red-edge, PSRI, and VARI) showed a statistically significant correlation with anthracnose incidence, indicating that the observed spectral variability primarily reflected the general canopy physiological condition rather than a disease-specific spectral response. These findings highlight both the potential and the current limitations of multispectral UAV imagery for anthracnose monitoring under tropical field conditions. Future research should incorporate multi-temporal UAV surveys, and complementary sensing technologies such as hyperspectral, thermal infrared, and LiDAR sensors, together with machine-learning approaches, to improve the early detection and spatial prediction of anthracnose in avocado production systems.

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